



IGA: isomorphism-aware graph alignment for robust unsupervised anchor link prediction

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Received: 5 June 2025 / Accepted: 20 September 2026

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Abstract

The proliferation of social platforms has led users to participate in multiple networks, giving rise to anchor links that connect accounts of the same individual across platforms. Anchor Link Prediction (ALP) seeks to uncover these links and underpins various downstream applications such as user profiling and cross-platform recommendation. While early methods rely on labelled anchor links for supervised training, real-world scenarios often lack such supervision, making unsupervised ALP both more practical and significantly more challenging. Existing unsupervised methods typically focus on learning from observed network data. However, they rarely take into account the generative processes underlying different networks. In this work, we propose an unsupervised framework, Isomorphism-aware Graph Alignment (IGA), based on the practical assumption that the source and target networks may contain regions that reflect shared structural information arising from partially overlapping network formation patterns, providing stable backbone components consistently observed across networks. IGA leverages the Graph Isomorphism Network (GIN) to align these backbone structures through a consistency-disparity learning principle. Additionally, we introduce a refinement strategy that adaptively augments structural and semantic information to improve alignment quality. Extensive experiments on real-world benchmarks demonstrate that IGA consistently outperforms existing unsupervised methods in both accuracy and robustness, particularly under noisy and sparse conditions.

Keywords Unsupervised anchor link prediction · Graph neural network · Backbone structure · Graph augmentation

1 Introduction

The proliferation of online platforms has led to the emergence of multiple independent networks, where users often maintain accounts across several services (Sun et al. 2012). Anchor Link Prediction (ALP), which aims to identify accounts for the same individual across different networks, plays a critical role in integrating fragmented information into a unified structure (Sun et al. 2019; Zhang et al. 2021). This integration reveals structural patterns that are otherwise hidden when networks are analyzed in isolation, thereby enabling downstream tasks such as user behavior modeling (Jiang et al. 2016), cross-platform recommendation (Shu et al. 2017; Zhang and Tong 2016), and protein-protein interaction alignment in bioinformatics (Faisal et al. 2015).

ALP is inherently challenging for two main reasons. First, only a small fraction of anchor links are typically available due to privacy concerns or users' reluctance to disclose identities across platforms (Backstrom et al. 2007; Fu et al. 2020). Second, real-world networks are often noisy, incomplete, and structurally heterogeneous (Wang et al. 2016), making it difficult to infer reliable alignments. These challenges are further intensified under unsupervised settings, as no anchor links are available to supervise training—leaving network topology as the sole signal for inference.

Supervised approaches address these difficulties by enforcing alignment consistency, requiring matched nodes to exhibit similar local structures across networks (Zhang and Yu 2015; Zhang and Tong 2016; Hadgu and Gundam 2020). They further incorporate disparity through negative sampling or heuristic labeling, ensuring that dissimilar nodes remain distinguishable in the embedding space (Liu et al. 2024). However, these techniques rely heavily on labeled anchor links, which are essential for both consistency enforcement and negative pair construction. In the absence of supervision, such methods cannot be directly applied. As a result, unsupervised ALP faces two core challenges: how to justify and exploit consistency without labeled pairs, and how to introduce meaningful disparity without access to negative examples. Existing unsupervised approaches that rely on independent embeddings or similarity heuristics often perform poorly under structural noise and lack principled mechanisms to enforce embedding separability.

Beyond methodological design, most unsupervised alignment methods seldom address the more fundamental question of why alignment is feasible in the first place. We argue that a theoretical understanding of the relationship between networks is crucial for principled unsupervised alignment. To this end, we assume that parts of the source and target networks can be viewed as realizations of a shared random process, providing backbone structures that guide alignment in a shared latent space.

Based on this assumption, we propose an unsupervised framework named LGA (Isomorphism-aware Graph Alignment). IGA enforces consistency by leveraging GCN's shared computational graph paradigm and the Weisfeiler-Lehman test. It promotes disparity by replacing the binary adjacency matrix with a modified Laplacian matrix as the reconstruction target, thereby helping prevent over-clustering and enhancing local separability. To further address noisy or incomplete links, IGA incorporates an adaptive graph augmentation strategy. By jointly refining the embedding

objective and the network structure, IGA provides a principled and robust solution for unsupervised anchor link prediction.

The main contributions of this paper are as follows:

- We introduce a practical assumption that parts of the source and target networks can be viewed as realizations from a shared random process, offering a principled foundation for unsupervised alignment.
- Based on this assumption, we develop a fully unsupervised framework that integrates consistency and disparity without requiring anchor links or negative examples.
- We propose a refinement strategy that adaptively augments the graph topology based to initial alignment results, surpassing previous methods that only modify the aggregation attention.
- Extensive experiments demonstrate that our approach achieves superior accuracy, robustness, and alignment quality under noisy and sparse conditions.

The remainder is organized as follows. Section 2 introduces our proposed method's preliminary, problem definition and notations. Section 3 provides the theoretical motivations behind our model design and the details of our proposed method. Section 4 reports the experimental results and discussion of the validity of our method. We review the related work in Sect. 5 and conclude our paper in Sect. 6.

2 Preliminary

2.1 Graph generation assumption

This work examines cross-platform user alignment, in which individuals may maintain accounts across multiple social networks. Although the observed networks differ, they may contain stable backbone regions that reflect recurring structural patterns, which can guide unsupervised alignment.

Let $\mathcal{V}$ denote the global set of potential users. For each observed graph G , a user $v_i \in \mathcal{V}$ has a conditional inclusion probability $p_i^{(G)} \in [0, 1]$, indicating the likelihood of v_i appearing in G . Each node is sampled independently, where v_i is included in G with probability $p_i^{(G)}$.

$$Z_i^{(G)} \sim \text{Bernoulli}(p_i^{(G)}), \quad V^{(G)} = \{v_i \in \mathcal{V} \mid Z_i^{(G)} = 1\}.$$

Conditional on the sampled node set $V^{(G)}$, edges are generated with conditional connection probabilities $q_{ij}^{(G)}$. Each entry $q_{ij}^{(G)} \in [0, 1]$ represents the likelihood of interaction between users v_i and v_j in Graph G . Edges are sampled as follows:

$$A_{ij}^{(G)} \sim \text{Bernoulli}(q_{ij}^{(G)}), \quad E^{(G)} = \{(v_i, v_j) \in V^{(G)} \times V^{(G)} \mid A_{ij}^{(G)} = 1\}.$$

This defines a distribution $\mathcal{G}(P^{(G)}, Q^{(G)})$ over each observed graphs, where $P^{(G)} = \{p_i^{(G)}\}$ and $Q^{(G)} = \{q_{ij}^{(G)}\}$ govern node and edge generation. The source and target networks are drawn according to their respective conditional distributions:

$$G_s \sim \mathcal{G}(P^{(G_s)}, Q^{(G_s)}), \quad G_t \sim \mathcal{G}(P^{(G_t)}, Q^{(G_t)}).$$

While the networks may differ overall, certain stable backbone nodes and edges correspond to components with similar conditional probabilities across G_s and G_t . These backbone components provide reliably observable signals that guide unsupervised alignment.

2.2 Backbone structure and alignment implication

To formalize recurring structures within the generative process, we introduce the concept of a latent backbone structure. The backbone comprises clusters of nodes and edges whose sampled appearances remain consistent across networks, offering reliable signals for alignment.

We begin with the node component. Let V_s and V_t denote the sets of sampled nodes in the source and target networks. The backbone node set is defined as the intersection of the two:

$$V_{\text{backbone}} = V_s \cap V_t.$$

Next, the backbone edge set consists of edges between backbone nodes that appear in both graphs:

$$E_{\text{backbone}} = \{(v_i, v_j) \in E_s \cap E_t \mid v_i, v_j \in V_{\text{backbone}}\}.$$

Together, these define the latent backbone graph:

$$G_{\text{backbone}} = (V_{\text{backbone}}, E_{\text{backbone}}).$$

The backbone can be interpreted as the portion of the networks *observed in both samples*. These overlapping nodes and edges provide structural signals that can guide unsupervised alignment, even when other parts of the networks differ or are partially missing.

Importantly, while these overlapping structures provide a useful basis for alignment, they are not directly observable and can be affected by sampling noise or structural variation. To address this challenge, we proposed a method that identifies and captures *clusters of backbone nodes and edges*, since isolated elements provide limited alignment information. Specifically, we employ Graph Neural Networks (GNNs) to aggregate local signals and highlight these cohesive backbone clusters, enabling robust unsupervised anchor link prediction.

3 Method

3.1 Notations

We denote the source and target networks as $G_s = (V_s, A_s, X_s)$ and $G_t = (V_t, A_t, X_t)$, where $V_s, V_t \subseteq \mathcal{V}$ are sampled node subsets from the global node pool. The adjacency matrices $A_s \in \{0, 1\}^{n_s \times n_s}$ and $A_t \in \{0, 1\}^{n_t \times n_t}$ correspond to edge sets E_s and E_t , with entries $a_{ij} = 1$ indicating the presence of an edge between nodes v_i and v_j . The node feature matrices $X_s \in \mathbb{R}^{n_s \times d}$ and $X_t \in \mathbb{R}^{n_t \times d}$ contain d -dimensional attribute vectors associated with each node.

An *anchor link* refers to a node pair (v_i, v'_j) such that $v_i \in V_s$, $v'_j \in V_t$, and both nodes correspond to the same latent entity. The goal of alignment is to identify such anchor links in an unsupervised setting.

We represent alignment likelihoods using a score matrix $S \in \mathbb{R}^{n_s \times n_t}$, where each entry s_{ij} quantifies the likelihood that $v_i \in V_s$ matches $v'_j \in V_t$. Anchor links can then be inferred by ranking entries in S (Heimann et al. 2018; Kollias et al. 2012).

Two core principles guide the alignment task. First, aligned nodes are expected to exhibit similar local structures, as their positions in the underlying latent graph, sampled from the same generative process, are assumed to be statistically correlated (Man et al. 2016). Second, aligned nodes often share a semantic similarity, as reflected in their node features (Kollias et al. 2012; Zhang and Tong 2016). Our method jointly leverages structural and attribute information to support alignment decisions.

These principles are naturally supported by our shared generation assumption, which provides a unified explanation for both topological and semantic dependencies across networks. Moreover, to address the challenges posed by noisy observations and partial sampling, we incorporate graph completion methods to enhance alignment robustness.

3.2 Motivation

Motivated by the characteristics of real-world datasets (Man et al. 2016) and the limitations of existing unsupervised methods (Heimann et al. 2018; Trung et al. 2020; Sun et al. 2023), we now introduce our approach, which aims to satisfy the following properties for a high-quality alignment model:

- *(P1) Consistency* Corresponding nodes across networks should have similar embeddings, ensuring that the alignment process reflects true structural and semantic similarities.
- *(P2) Disparity* Dissimilar nodes within the same network should remain distinguishable in the embedding space to prevent erroneous matches and preserve network identity.
- *(P3) Robustness* The model should tolerate structural discrepancies caused by randomness, sparsity, or missing data and remain effective under real-world noise.

To meet these goals, our proposed IGA framework integrates the following key designs:

To encourage (P1) consistency, we start from the ideal case where the source and target graphs are structurally similar. In such settings, the Weisfeiler-Lehman (WL) test (Cai et al. 1992; Douglas 2011) can effectively identify aligned nodes. However, real-world graphs rarely satisfy the isomorphism condition. We thus adopt the Graph Isomorphism Network (GIN) (Xu et al. 2019), which retains WL's expressive power while operating under noisy conditions. Using GIN with shared parameters, we ensure structurally similar nodes are embedded close, facilitating consistent alignment.

To promote (P2) disparity, we replace the binary adjacency matrix with the graph Laplacian during training. This enhances sensitivity to finer topological differences, avoids over-smoothing, and allowing distinct nodes to maintain separation, reducing the risk of false matches.

To improve (P3) robustness, we introduce a refinement phase combining structural augmentation with semantic reinforcement. Augmenting edges alleviates sparsity and structural incompleteness while reinforcing confidently aligned pairs, strengthening representation. These mechanisms jointly improve the model's resilience and alignment accuracy.

An overview of the IGA framework is shown in Fig. 1.

3.3 Assurance of consistency

In the ideal case where the source and target networks are isomorphic, corresponding nodes share identical local structures. Under such conditions, the Weisfeiler-Lehman

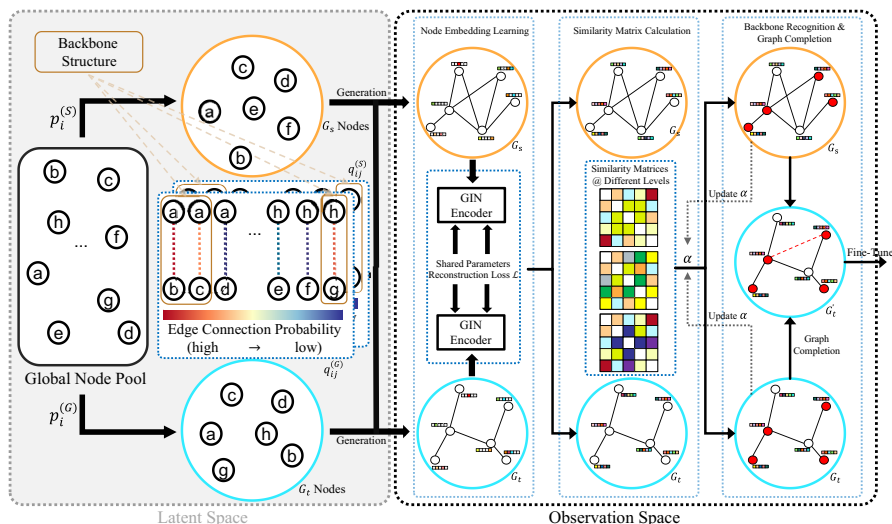


Fig. 1 Illustration of the proposed framework. Source and target networks are independently sampled from the global node pool. Backbone clusters, consisting of nodes and edges observed in both graphs, are encoded via a shared GIN encoder. A reconstruction loss maintains structural disparity, while graph completion and fine-tuning enhance alignment accuracy

(WL) graph isomorphism test (Cai et al. 1992; Douglas 2011) assigns identical labels to aligned nodes through iterative neighborhood aggregation:

$$\ell^{(k+1)}(v) = \text{HASH}(\ell^{(k)}(v), \{\ell^{(k)}(u) : u \in N(v)\}),$$

where $N(v)$ denotes the neighbors of v , and HASH is an injective function (Bakhtiari et al. 1995). This forms the theoretical basis for structural consistency.

Proposition 1 *If source network G_s and target network G_t are isomorphic, then corresponding nodes of anchor links will have the identical label after K iterations of Weisfeiler-Lehman Relabeling.*

However, identical labels can still emerge among distinct nodes even with uniform initialization. Figure 2 (third group, left to right) demonstrates that each subgraph contains two node pairs sharing labels after relabeling. This blocks further structural matching due to insufficient distinguishing information. Identical initial labels fail to convey node-level diversity. To remedy this, we embed node-specific features X at initialization, supplying richer information and enabling the model to differentiate structurally similar yet semantically distinct nodes. This motivates tracking the propagation of structural and semantic information through computational graphs during aggregation. Figure 3 illustrates the identical computation trees that arise in this setting.

Lemma 1 *Initializing the Weisfeiler-Lehman (WL) relabeling process with node-specific feature vectors increases its expressive power in distinguishing non-identical nodes within the same graph.*

Figure 4 shows how feature-informed initialization breaks the ambiguity between the two computation trees.

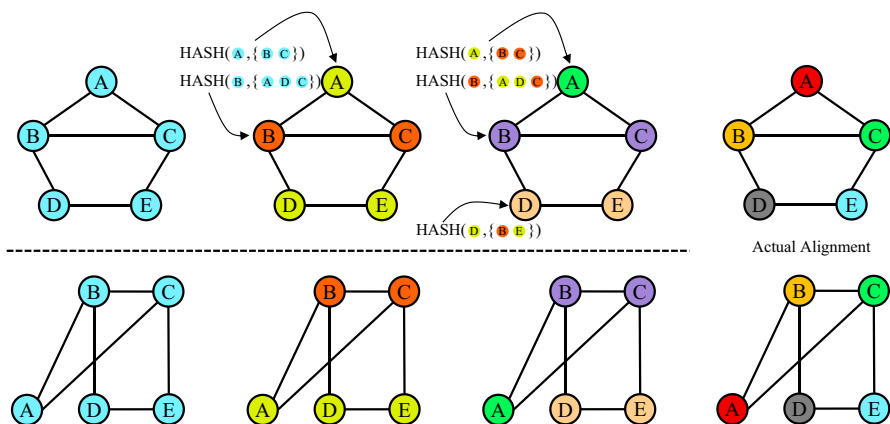


Fig. 2 Illustration of the Weisfeiler-Lehman relabeling process and its connection to anchor link prediction

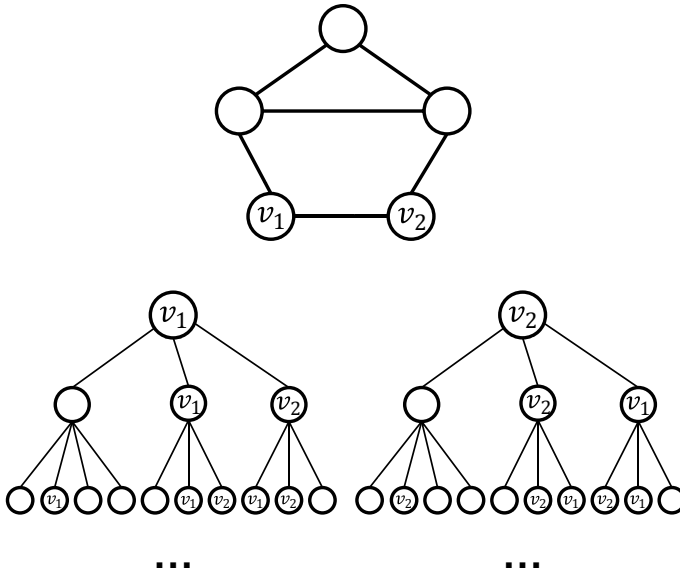


Fig. 3 Computational graphs of v_1 and v_2 with identical local structures

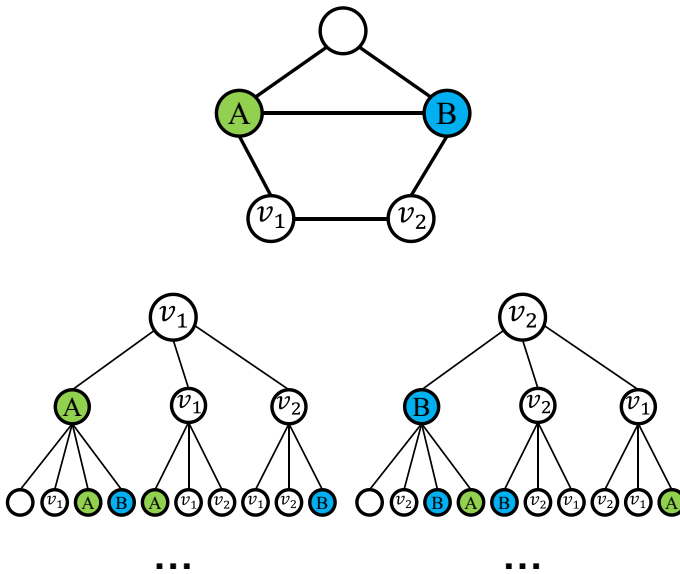


Fig. 4 Feature-informed computational graphs of v_1 and v_2 enable finer discrimination

To handle real-world deviations from isomorphism, we relax the WL test in two ways: (i) limiting its depth to rely on stable local neighborhoods, and (ii) replacing the discrete HASH function with a continuous, learnable aggregator. We adopt the Graph Isomorphism Network (GIN) (Xu et al. 2019), which retains WL-level expressiveness while producing continuous embeddings.

GIN encodes each node as follows:

$$H_v^{(0)} = X(v), \quad (1)$$

$$H_v^{(l)} = \text{MLP} \left((1 + \epsilon) H_v^{(l-1)} + \sum_{u \in \mathcal{N}(v)} H_u^{(l-1)} \right), \quad (2)$$

where ϵ is a learnable parameter and $\text{MLP}(\cdot)$ denotes a multi-layer perceptron.

Proposition 2 *If two nodes $v \in G_s$, $v' \in G_t$ share isomorphic neighborhoods and identical features under recursive matching, then $H_s^{(l)}(v) = H_t^{(l)}(v')$ for any l under shared GIN parameters.*

We train GIN under an unsupervised autoencoder framework by reconstructing a structure-aware target matrix Y :

$$\mathcal{L} = \|Y - H'^{(l)} H'^{(l)T}\|_F^2, \quad (3)$$

where $H'^{(l)}$ is the normalized embedding. This approach allows the model to learn informative representations solely from the internal structure of each network without relying on anchor supervision. Cross-network similarity is computed via:

$$S^{(l)} = H_s'^{(l)} H_t'^{(l)T}, \quad (4)$$

providing alignment matrices at multiple GIN layers to capture structural similarity at varying receptive field sizes.

Notably, the choice of Y plays a critical role: it not only guides unsupervised embedding learning but also directly affects how intra-network node relationships are encoded. In particular, our specific design of Y introduces inductive bias into the model that helps preserve node separation, an aspect we further exploit in the next section to promote embedding disparity.

3.4 Assurance of disparity

In anchor link prediction, node embeddings must remain well-separated in the latent space (Zhang et al. 2021) so that valid anchor pairs can be effectively distinguished

from unrelated nodes. A key challenge arises when structurally similar but semantically different nodes within the same graph are embedded too closely, commonly referred to as hard negatives (Robinson et al. 2021). This phenomenon is particularly problematic in unsupervised settings and significantly hinders alignment accuracy.

Standard autoencoders often reconstruct the adjacency matrix A (Kipf and Welling 2016), thereby encouraging local proximity in the embedding space. Although this preserves neighborhood information, it may unintentionally cause adjacency collapse: excessive similarity among neighboring nodes, even when structurally similar but semantically unrelated. As a result, these nodes may collapse into nearby representations, leading to more false-positive matches.

Proposition 3 *Training to reconstruct the adjacency matrix can introduce hard negatives, degrading anchor link prediction performance.*

Beyond this, adjacency-based objectives may also cause feature collapse (Trung et al. 2020), where embeddings lose variance and become indistinguishable. To address this, we propose reconstructing the graph Laplacian, which provides a more global, degree-sensitive view of the graph. This promotes better node separation and preserves richer structural cues.

Lemma 2 *Using the normalized Laplacian as reconstruction target mitigates the formation of hard negatives in the downstream alignment task.*

To further enhance stability, we adopt a renormalized Laplacian $\hat{L}$ following (Kipf and Welling 2017), where:

$$\hat{L}(u, v) = \frac{1}{\sqrt{(d_u + 1)(d_v + 1)}}.$$

This formulation produces small, non-zero weights for connected nodes (e.g., 0.5 for degree-one pairs), maintaining smoothness while enhancing disparity.

Accordingly, we revise the loss function as:

$$\mathcal{L} = \|\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} - H'^{(L)} H'^{(L)T}\|_F^2, \quad (5)$$

where $\hat{A} = A + I$, and $\hat{D}_{ii} = \sum_j \hat{A}_{ij}$ is the degree matrix. The output $H'^{(L)}$ is the normalized embedding from the final GIN layer.

By reconstructing the renormalized Laplacian, the model yields a more dispersed embedding space that fulfills the node-level disparity constraint.

Having tackled disparity via embedding space dispersion, we now pivot to the final property: resilience to structural and feature-level perturbations.

3.5 Assurance of robustness

Although the source and target networks are distinct, locally cohesive regions reveal partially shared information from network formation. These regions do not necessarily indicate a unified generative process; rather, they pinpoint zones where structural patterns and semantic attributes converge. Backbone nodes are distinguished by their topological and semantic integrity, offering compelling evidence for alignment. As Proposition 2 suggests, nodes with aligned structures and attributes yield analogous embeddings under a shared GIN encoder. This property permits reliable extraction of anchor links, as detailed below.

3.5.1 Identifying backbone structure

We first pinpoint high-confidence nodes that consistently demonstrate robust embedding similarity across GIN layers. To harness multi-scale topological signals, we fuse similarity matrices from all K layers:

$$S = \sum_{l=0}^K S^{(l)} = \sum_{l=0}^K H_s'^{(l)} H_t'^{(l)T}, \quad (6)$$

where $H_s'^{(l)}$ and $H_t'^{(l)}$ are the normalized embeddings at layer l . A node pair (v, v') is selected as a backbone match if:

$$\begin{cases} \operatorname{argmax}_{u' \in V_t} S^{(l)}(v, u') = v' & \forall 0 \leq l \leq K, \\ \operatorname{argmax}_{u \in V_s} S^{(l)}(u, v') = v & \forall 0 \leq l \leq K, \\ S^{(l)}(v, v') > \lambda & \forall 0 \leq l \leq K. \end{cases} \quad (7)$$

To strengthen alignment, we use backbone anchor pairs that are distinguished by structural and semantic integrity. Amplifying the role of these stable nodes ensures their influence spreads efficiently, enabling the identification of additional aligned pairs even under noisy, incomplete conditions. This strategic shift improves robustness and reliability throughout the alignment framework.

3.5.2 Backbone-guided expansion

We then intensify the influence of these backbone nodes to steer subsequent alignment. A weight $\alpha(v)$ is assigned to each node and updated via:

$$\alpha(v) \leftarrow \alpha(v) \cdot \beta, \quad (8)$$

leading to the revised GIN message-passing rule:

$$H_v^{(l)} = \text{MLP} \left((1 + \epsilon) \alpha(v) H_v^{(l-1)} + \sum_{u \in \mathcal{N}(v)} \alpha(u) H_u^{(l-1)} \right), \quad (9)$$

This weighting strategy strengthens nodes that are more semantically and structurally stable, thereby increasing robustness in information propagation during aggregation. This approach, while distinct in implementation, mirrors the spirit of the Graph Information Bottleneck (GIB) principle (Wu et al. 2020), which preserves downstream-relevant information by filtering out redundant or noisy signals.

3.5.3 Backbone structure based graph completion

Backbone anchor pairs are expected to give shared structural cues. However, their local neighborhoods may still differ between the two observed networks. Discrepancies can come from incomplete observations, platform-specific interactions, or noisy edges, all of which may weaken message propagation across matched nodes. To address these challenges and improve robustness, we add a graph-completion step that restores alignment within the backbone by synchronizing edges across matched node pairs. This ensures consistent connectivity for reliable information flow:

$$(v, u) \in E_s \iff (v', u') \in E_t, \quad (10)$$

for stable matches (v, v') and (u, u') . This operation restores structural consistency across graphs by enforcing symmetric edge connectivity within the backbone.

3.5.4 Fine-tuning on augmented graphs

Since the GIN model is trained via Laplacian reconstruction (Eq. 5), modifying the adjacency structure requires fine-tuning. This adjustment aligns embeddings with the updated topology while preserving previous knowledge. By integrating graph refinement with embedding adaptation, the model achieves stable and robust performance under noisy or incomplete network conditions, thereby satisfying the robustness property.

Having discussed all components, we now transition to a summary of the overall algorithm.

Require: Source graph $G_s = (V_s, \mathbf{A}_s, \mathbf{X}_s)$, target graph $G_t = (V_t, \mathbf{A}_t, \mathbf{X}_t)$, amplification factor β

Ensure: Node alignment matrix $\mathbf{S}$

- 1: Initialize GIN encoder f_θ (including ϵ and MLP parameters)
- 2: Initialize node weights $\alpha(v) = 1$ for all v
- 3: Initialize backbone set $B = \emptyset$

Training Pipeline:

- 4: Pre-train f_θ on initial Laplacians $\hat{L}_s, \hat{L}_t$
- 5: **repeat**
- 6: Compute node embeddings:
- 7: $\mathbf{H}_s = f_\theta(G_s, \alpha)$
- 8: $\mathbf{H}_t = f_\theta(G_t, \alpha)$
- 9: Compute similarity matrix $\mathbf{S} = \mathbf{H}_s \mathbf{H}_t^T$
- 10: Select backbone matches B' using Eq. (7)
- 11: Update node weights $\alpha(v)$ for $v \in B'$ using Eq. (8)
- 12: Apply graph completion (Eq. (10)) to G_s and G_t
- 13: Fine-tune f_θ on updated Laplacians
- 14: Update backbone set $B \leftarrow B \cup B'$
- 15: **until** No new backbone matches are found

Inference Procedure:

- 16: For new or unaligned nodes:
- 17: Compute embeddings $\mathbf{H}_s, \mathbf{H}_t = f_\theta(G_s, G_t, \alpha)$
- 18: Compute similarity matrix $\mathbf{S} = \mathbf{H}_s \mathbf{H}_t^T$
- 19: Determine final alignment from $\mathbf{S}$
- 20: **return** $\mathbf{S}$

Algorithm 1 IGA: GIN-based unsupervised anchor-link prediction (training and inference)

3.6 Complexity analysis

Let N and E denote the number of nodes and edges in a network. The main components are summarized below:

- *GIN embedding computation:*
 - Each GIN layer aggregates information from a node's neighbors and applies a linear transformation followed by a nonlinear activation.
 - Aggregation requires $O(N + E)$ operations per layer, and the MLP transformation requires $O(N \cdot d^2)$ operations, where d is the embedding dimension.
 - With L layers, the total time complexity is $O(L \cdot (N + E) \cdot d + L \cdot N \cdot d^2)$.
 - Space complexity is $O(N \cdot d)$ to store node embeddings at each layer.

- *Similarity matrix computation:*
 - Computing the full pairwise similarity between source and target embeddings requires $O(N^2 \cdot d)$ operations.
 - Storing the similarity matrix requires $O(N^2)$ memory.
- *Backbone selection and graph completion:*
 - Selecting backbone matches involves checking candidate node pairs, costing $O(N^2)$ time.
 - Updating edges during graph completion requires $O(E)$ time.
 - Additional memory beyond adjacency matrices is negligible.

To contextualize these estimated costs, we now summarize the main computational components and their scaling behavior. The main computational costs are node embedding computation and similarity matrix construction. These estimates provide practical guidelines for understanding the method's computational demands and how it scales with network size, especially given the lack of large-scale benchmark datasets for unsupervised anchor link prediction.

4 Experiments

In the experiment section, we aim to answer the following research questions.

- (RQ1): Does our model outperform existing baseline methods?
- (RQ2): Is our model robust to structural noise?
- (RQ3): Does the GIN architecture provide a significant improvement over vanilla GCN?
- (RQ4): Does the fine-tuning stage lead to meaningful performance gains?
- (RQ5): Does using the normalized Laplacian in the loss function yield a notable difference?
- (RQ6): What structural patterns can be observed in the identified backbone, and do they provide exploratory evidence for the practical assumptions behind our alignment framework?

In the remainder, we describe the experimental setup, including datasets, baseline methods, and evaluation metrics. Then, we present our empirical results, robustness against structural noise, and ablation studies.

4.1 Experimental setup

Datasets We conduct experiments using both real-world and synthetic network alignment datasets. Our real-world datasets consist of two alignment benchmarks: Douban Online vs. Douban Offline and Allmovie vs. Imdb. These cover social and collaboration networks from different domains. For synthetic datasets, we use cita-

tion networks (Cora and Citeseer) from the Planetoid collection. We generate source and target networks by randomly removing edges to simulate structural noise while preserving node identities. Node features are reduced to 20 dimensions using truncated SVD (Halko et al. 2011). Full dataset descriptions and statistical details appear in Appendix 1.

4.1.1 Baseline methods

We compare six methods, grouped by their use of topological information and node attributes. The first group uses only topological information: IsoRank (Singh et al. 2008), PALE (Man et al. 2016), and CENALP Du et al. (2019). The second uses both topological information and node attributes: REGAL (Heimann et al. 2018), GAlign (Trung et al. 2020), and HTC (Sun et al. 2023). GAlign (Trung et al. 2020) and HTC (Sun et al. 2023) are most similar to our approach as both use unsupervised network alignment with Graph Convolutional Networks (GCNs). Some baseline methods, such as PALE and CENALP, require supervised training, using 10% of the ground truth. We also create prior alignment matrices for IsoRank.

4.1.2 Evaluation metrics

Two widely used numerical metrics are used here to evaluate and compare the effectiveness of different approaches. The first one is *precision@k* (Zhang and Tong 2016), which measures the ratio of a node's true anchor counterpart and has a similarity value that lies among the top k candidates. Mathematically, *precision@k* is defined as:

$$\text{precision@k} = \frac{\sum_{v_s^* \in V_s^*} \mathbb{I}(\text{rank}(v_t^*) \leq k)}{|V_s^*|}, \quad (11)$$

where (v_s^*, v_t^*) is an anchor link, V_s^*, V_t^* stands for the anchor nodes set for source and target networks respectively, $\mathbb{I}(\cdot)$ is an indicator function, and S is the similarity matrix.

The second metric is called *Mean Average Precision* (MAP) (Man et al. 2016), and it evaluates the average rank of the first correct prediction. It is defined as follows:

$$\text{MAP} = \frac{1}{|V_s^*|} \sum_{v_s^* \in V_s^*} \frac{1}{r_{v_s^*}}, \quad (12)$$

where $r_{v_s^*}$ stands for the rank of the actual anchor node's alignment score among its candidates. Note that the higher the values are, the better performance for both *precision@k* and MAP.

4.1.3 Configuration of experiment

Unless specified, hyperparameters are set as follows: number of GIN layers $L = 2$, embedding dimension $d = 200$, learning rate $\eta = 0.005$, importance amplifier $\beta = 1.1$

. All results are reported as mean $\pm$ standard deviation over 20 runs to mitigate randomness. For baseline, we follow the hyperparameters as recommended in their original papers. GAlign and HTC use the same datasets and evaluation protocol as IGA. Embedding-based methods are set to $d = 200$ for fair comparison. For classical baselines such as PALE, CENALP, IsoRank, and REGAL, we use standard benchmark configurations, including 10% ground-truth anchor links for PALE and CENALP and prior alignment matrices for IsoRank.

4.2 Result and analysis

4.2.1 Overall performance

Table 1 answers RQ1 by presenting a comprehensive comparison between IGA and several representative baselines across four benchmark datasets. For clarity, values are scaled by 100. Overall, IGA shows strong performance across all datasets and evaluation metrics.

On the Allmovie & Imdb dataset, IGA achieves the best *precision@10* (92.23) and MAP (86.02) while attaining a competitive *precision@1* (82.96), slightly trailing HTC. This result indicates that IGA is highly effective at identifying a broad set of top-ranked candidate matches, even if the top-1 prediction is occasionally surpassed.

On the Douban Online & Offline dataset, which features a sparser structure and lower average degree, IGA clearly outperforms all baselines in *precision@1* (49.11) and MAP (60.15). The performance gap between IGA and the strongest baseline (HTC at 46.51 for *p@1*) highlights IGA's robustness in aligning low-degree nodes and handling structurally incomplete graphs.

To evaluate our model under settings that strictly conform to the assumption that both networks are generated from the same underlying random process, we construct two synthetic datasets, Cora 40% and CiteSeer 40%. Each dataset is formed by independently removing 40% of edges from a single original graph to obtain two partially observed views, with the node set kept identical. This design naturally aligns with our generation assumption and enables us to examine alignment quality under stochastic structural divergence. On both datasets, IGA outperforms all baselines across *p@1*, *p@10*, and MAP. Notably, IGA achieves *p@1* of 91.14 on Cora and 92.00 on CiteSeer, reflecting its ability to recover consistent alignments even under significant random structural variation.

These results collectively demonstrate that IGA not only surpasses traditional methods in real-world settings but also maintains strong alignment accuracy under unsupervised and assumption-conforming scenarios, highlighting its adaptability to diverse and noisy network conditions.

4.2.2 Robustness test

We address RQ2 by evaluating the robustness of our model under varying levels of structural noise. Following the same assumption that source and target networks are independently sampled from a shared underlying graph, we generate two views of each dataset (Cora and CiteSeer) by randomly deleting 10–50% of edges from the

Table 1 Network alignment performance

(a) Real-world datasets

Method	Allmovie and Imdb			Douban online and offline		
	$p@1$	$p@10$	Time(s)	$p@1$	$p@10$	Time(s)
IGA	82.96 ± 0.68	92.23 ± 0.51	138.47	49.11 ± 0.84	79.53 ± 0.57	60.15 ± 0.57
HTC	84.36 ± 0.88	90.51 ± 0.39	108.57	46.51 ± 0.85	80.05 ± 0.34	57.66 ± 0.67
GAlign	82.14 ± 0.71	90.03 ± 0.33	135.48	45.26 ± 0.67	78.00 ± 0.54	56.32 ± 0.55
PALE	69.47 ± 0.98	71.59 ± 0.42	2015.42	7.75 ± 0.68	44.79 ± 0.56	19.01 ± 0.72
CENALP	48.66 ± 0.45	83.27 ± 0.32	42786.61	25.72 ± 0.39	46.18 ± 0.42	35.37 ± 0.47
IsoRank	46.53 ± 0.47	64.27 ± 0.32	142.41	9.03 ± 0.40	20.48 ± 0.36	12.99 ± 0.38
REGAL	9.53 ± 0.64	38.69 ± 0.52	156.78	5.28 ± 0.41	23.26 ± 0.53	11.20 ± 0.39

(b) Synthetic dataset

Method	Cora 40%			CiteSeer 40%		
	$p@1$	$p@10$	Time(s)	$p@1$	$p@10$	Time(s)
IGA	91.14 ± 0.17	99.52 ± 0.07	13.79	92.00 ± 0.23	99.76 ± 0.06	95.14 ± 0.23
HTC	90.02 ± 0.13	99.23 ± 0.06	12.82	90.45 ± 0.16	99.34 ± 0.07	95.26 ± 0.13
GAlign	89.96 ± 0.22	99.00 ± 0.11	<u>13.52</u>	89.66 ± 0.26	99.20 ± 0.08	94.82 ± 0.25
PALE	72.31 ± 0.26	88.92 ± 0.13	54.28	70.74 ± 0.34	87.25 ± 0.24	79.56 ± 0.29
CENALP	61.28 ± 0.17	86.71 ± 0.09	10975.84	59.02 ± 0.23	85.09 ± 0.12	69.35 ± 0.26
IsoRank	40.27 ± 0.23	68.80 ± 0.13	29.82	38.45 ± 0.21	67.05 ± 0.14	50.11 ± 0.21
REGAL	10.56 ± 0.19	49.81 ± 0.12	28.78	9.94 ± 0.16	48.33 ± 0.18	22.35 ± 0.27

Values are scaled by 100 and reported as mean $\pm$ std over 20 runs. Best results are in bold and second-best are underlined

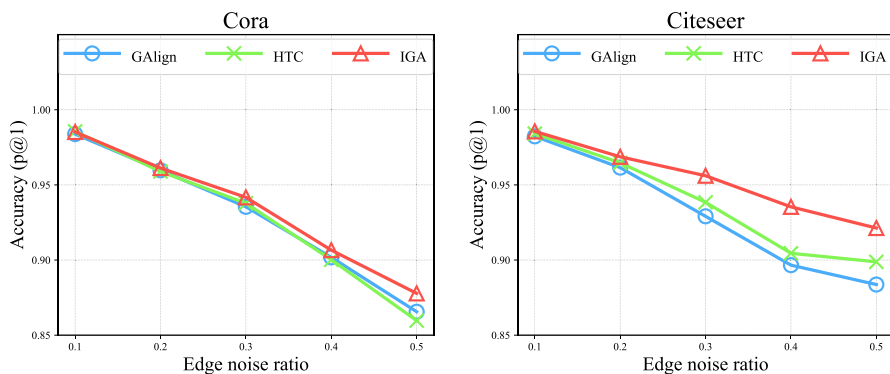


Fig. 5 Robustness test against structural noise (Cora and CiteSeer)

original graph. This setting introduces controlled structural divergence while maintaining the same node set, simulating realistic yet assumption-aligned variations.

Figure 5 reports the *precision@1* performance of IGA compared to two competitive baselines, GALign and HTC. We focus on these two due to their state-of-the-art performance, as prior studies (Trung et al. 2020; Sun et al. 2023) suggest other baselines fall significantly behind in noisy or unsupervised scenarios.

Across all levels of edge removal, IGA maintains consistent superiority over the baselines. At low noise levels (e.g., 10% deletion), the performance gap is minor, with all methods achieving *p@1* near 0.98. However, as noise increases, IGA exhibits significantly better resilience. For example, at 50% edge deletion, IGA retains *p@1* scores of 0.8778 (Cora) and 0.9213 (CiteSeer), whereas both HTC and GALign show larger performance degradation. In addition to achieving higher absolute accuracy, IGA demonstrates better stability, with a smoother degradation trend as noise increases.

These results validate IGA’s robustness to structural inconsistencies and highlight its ability to preserve alignment quality even under substantial random perturbations.

4.2.3 Ablation study

To better understand the contribution of each core component in our model, we conduct an ablation study addressing three questions: RQ3 whether the GIN architecture brings substantial advantages over simpler alternatives; RQ4 whether the fine-tuning stage contributes significantly to performance; and RQ5 whether using the normalized Laplacian as input provides benefits over the adjacency matrix.

Table 2 summarizes the results of six model variants evaluated on the Allmovie & Imdb and Douban Online & Offline datasets. The full IGA model combines GIN encoding, normalized Laplacian input, graph completion, and fine-tuning. Variants are constructed by selectively removing or replacing specific components to isolate their effects.

Compared to GALign, our GIN-based models consistently achieve higher performance across all metrics, confirming the expressiveness of the GIN encoder RQ3. Notably, even without fine-tuning, IGA outperforms GALign by a large margin, indi-

Table 2 Ablation study results across different model configurations

Method	Fine-tuning	Encoder input	Graph completion	Allmovie and Imdb		Douban on and off	
				$p@1$	MAP	$p@1$	MAP
IGA	✓	Modified Laplacian	✓	0.8296	0.8602	0.4911	0.6015
IGA-1	✓	Adjacency	✓	0.5879	0.6730	0.2879	0.3784
IGA-2	✓	Original Laplacian	✓	0.7942	0.8417	0.4451	0.5378
IGA w/o Fine-tuning	✗	Modified Laplacian	✗	0.7128	0.7773	0.4821	0.5890
IGA w/o GC	✓	Modified Laplacian	✗	0.8145	0.8234	0.4745	0.5547
GAlign w/o FT	✗	Modified Laplacian	✗	0.5458	0.6386	0.4320	0.5419

Table 3 Robustness test without graph completion in the fine-tuning process

Method	CiteSeer ($p@1(\uparrow)$)				
	10%	20%	30%	40%	50%
IGA	0.9856	0.9687	0.9561	0.9354	0.9213
IGA w/o GC	0.9834	0.9627	0.9384	0.9175	0.8972
GAlign	0.9823	0.9615	0.9291	0.8966	0.8837

cating that the core encoder architecture contributes significantly to structural awareness. Additionally, when the normalized Laplacian is replaced with a plain adjacency matrix (IGA-1), performance drops substantially, especially on the sparser Douban dataset. This supports our hypothesis that the Laplacian better preserves structural disparity needed for reliable unsupervised alignment RQ5. The intermediate variant IGA-2, which uses the Laplacian but omits graph completion and fine-tuning, performs better than IGA-1 but still falls short of the complete model.

The importance of the fine-tuning stage is confirmed by comparing IGA w/o FT and the full IGA model RQ4. Fine-tuning yields consistent gains across both datasets, particularly in MAP, suggesting improved ranking quality for candidate alignments. Furthermore, including graph completion during fine-tuning yields additional gains in alignment accuracy. IGA w/o GC performs worse than the full model by 0.0151 and 0.0468 in $p@1$ on Allmovie and Douban, respectively, showing that restoring high-probability links between aligned nodes mitigates structural inconsistencies and strengthens match reliability.

These results collectively demonstrate that each component of IGA contributes meaningfully to the final performance. GIN encoding and Laplacian input enhance representational capacity while fine-tuning and graph completion stabilize alignment quality in the presence of incomplete or noisy structures.

To further quantify the benefit of graph completion under structural noise, we also evaluate performance on the Citeseer dataset with varying edge-deletion ratios. Table 3 shows that as the deletion ratio increases from 10% to 50%, all methods experience a degradation in performance. However, IGA maintains a clear lead throughout and degrades more smoothly than its counterparts. For instance, at 50% edge deletion, the $p@1$ score of IGA remains at 0.9213, while IGA w/o GC drops to 0.8972 and GAlign drops further to 0.8837. These results highlight the robustness of our fine-tuning strategy and confirm the effectiveness of structural completion in preserving alignment quality under noisy conditions.

4.2.4 Validity of the shared generation assumption

This section addresses RQ6, which explores the structural evidence observed in the identified backbone and how this evidence supports the practical assumptions underlying our alignment framework. We do not aim to fully validate the shared-generation assumption based solely on degree distributions. Instead, we use this analysis as an exploratory check to understand whether the identified backbone exhibits meaningful structural preferences.

We visualize the degree distributions of the original source and target networks, along with their corresponding identified backbones, in Fig. 6. The figure presents four subplots arranged in a 2×2 grid. The top row shows the degree distributions (histogram and empirical PDF) of the original source and target graphs, while the bottom row shows the degree distributions of the corresponding backbone nodes.

The main observation comes from comparing each original graph with its corresponding backbone. Compared with the original networks, the backbone distribu-

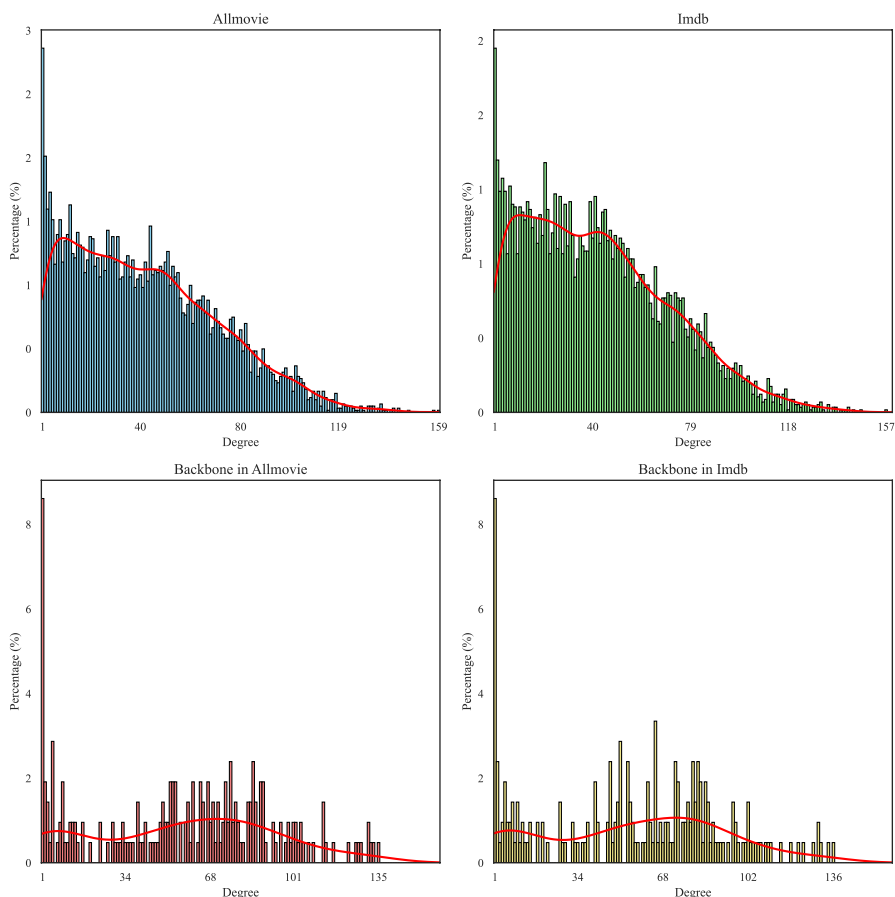


Fig. 6 Degree distributions (histogram + empirical PDF) for original networks and their identified backbones. Top: source and target graphs; bottom: corresponding backbones

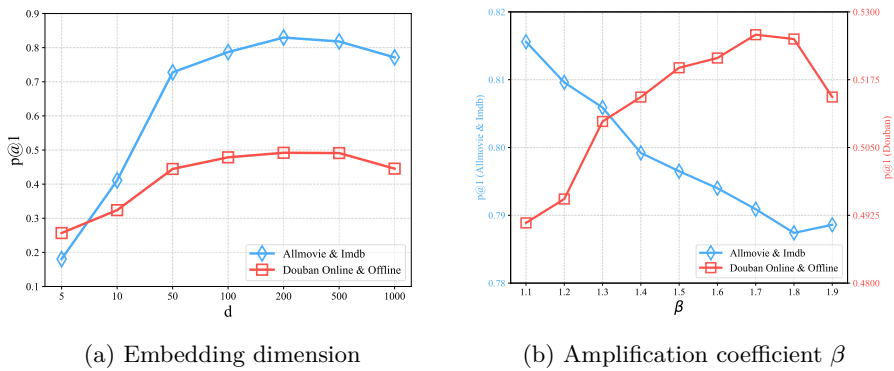


Fig. 7 Key hyperparameters' influence on $p@l$

Table 4 Hyperparameter study on number of GIN layers

Layers	Allmovie and Imdb		Douban online and offline	
	$p@l$	MAP	$p@l$	MAP
1	0.6782	0.7042	0.3785	0.4255
2	0.8296	0.8602	0.4911	0.6015
3	0.8044	0.8459	0.4844	0.5969
4	0.7905	0.8324	0.4410	0.5532

tions are shifted toward nodes with relatively larger degrees. This suggests that the identified backbone is not formed by arbitrarily scattered nodes. Instead, it tends to concentrate around more connected regions of the networks. These observations are consistent with our motivation that useful backbone structures should appear as locally cohesive clusters rather than isolated nodes or edges. Although degree distribution alone cannot prove the underlying generative assumption, it provides supportive evidence that the model can identify structurally meaningful backbone regions. These regions then serve as practical cues for unsupervised alignment.

4.2.5 Hyperparameter study

We conduct a series of experiments to analyze IGA's sensitivity to several key hyperparameters, including the number of GIN layers, the embedding dimension, and the amplification coefficient β . Figure 7 summarizes the embedding-dimension and amplification-coefficient results.

First, we examine the impact of the number of GIN layers (L). Prior studies (Xu et al. 2018; Zhang and Chen 2018) have suggested that deeper GCN-like models may suffer from over-smoothing beyond two layers. As shown in Table 4, our model also achieves the best performance with two GIN layers. Performance degrades slightly when increasing to three layers, and more significantly at four, indicating that shallow architectures are more effective for this task.

Next, we analyze the effect of the embedding dimension (d). As illustrated in Fig. 7a, increasing d from small values significantly improves performance up to around 200 dimensions. Beyond that, the gains diminish, and performance drops

when $d = 1000$, likely due to overfitting. Based on this, we set $d = 200$ for all experiments to balance expressiveness and generalization.

Finally, we study the influence of the amplification coefficient β , which modulates the structural importance signal. As shown in Fig. 7b, the Allmovie & Imdb dataset performs best under smaller β values, while the Douban dataset benefits from a larger β around 1.7. Divergence suggests that noisier networks require stronger structural emphasis to guide effective alignment.

5 Related work

5.1 Network alignment

Recently, Network Embedding (NE) offered a promising avenue for acquiring latent low-dimensional embedding about each node in noisy and sparse situations by aggregating their “higher-order” neighbors to learn discriminative representations of nodes (Grover and Leskovec 2016; Hamilton et al. 2017; Spinelli et al. 2021). NE-based ALP methods aim to infer node alignments using low-dimensional node embeddings. Many NE-based ALP methods, such as M-NASA (Zhang and Yu 2015), FINAL (Zhang and Tong 2016), and MC (Yang et al. 2022), learn node embeddings under the assumption of alignment consistency across different networks. That is, the same node maintains a consistent connectivity structure across different networks (Mu et al. 2016; Hadgu and Gundam 2020; Yang et al. 2022), which is supported by evidence that many of these networks share common structures due to similar network features (Zhang and Yu 2015; Zhan et al. 2017). For instance, Zhang et al. (Zhang and Tong 2016) explicitly designated alignment consistency as the minimization of differences between the alignment of two nodes and that of their respective neighbors in FINAL. However, despite the powerful embedding-learning ability of NE methods, the pursuit of consistency may exacerbate over-smoothing within local neighborhoods and hinder the differentiation between accurate and misleading alignments (Liu et al. 2024). Moreover, real-world scenarios often violate the assumption of alignment consistency, as it is commonly observed that a user’s behavior across different social networks can be highly diverse (Chu et al. 2019; Shi et al. 2022).

5.1.1 Supervised approach

Classical network alignment methods often rely on supervised learning, where supervisory information is provided by partial ground truth (e.g., PALE (Man et al. 2016), CENALP (Du et al. 2019)) or prior alignment matrices (e.g., IsoRank (Singh et al. 2008), FINAL (Zhang and Tong 2016)). These methods aim to reconstruct embedding spaces using external labels as a basis. However, they suffer from the labor-intensive nature of manual labeling, which limits their practicality for large-scale or real-world applications. In contrast, unsupervised methods, which do not require labeled data, are increasingly favored due to their flexibility and applicability to large-scale networks.

5.1.2 Unsupervised approach

Traditional methods often rely on string distance metrics (such as TF-IDF, Jaccard, and Euclidean distance) to compare node names or descriptions (Otero-Cerdeira et al. 2015). However, these approaches are vulnerable to textual noise and do not utilize any topological structure of the network. With the development of network embedding techniques (Goyal and Ferrara 2018), especially graph convolutional networks (Kipf and Welling 2017), GCN-based unsupervised network alignment methods have been proposed recently. GAlign (Trung et al. 2020) employs a multi-order embedding model that maintains consistency with graph convolutional neural representations. HTC (Sun et al. 2023) incorporates higher-order topological information, namely edge orbits, to train embeddings under alignment consistency. Our work goes beyond these state-of-the-art methods by providing theoretical support for model selection and incorporating topological enhancement in fine-tuning.

5.2 Graph convolutional network

Graph Convolutional Network (GCN) was introduced to address the challenge of learning from graph-structured data by extending the convolutional neural network (CNN) to non-Euclidean domains. Initially proposed by Kipf and Welling in 2016 (Kipf and Welling 2017), GCN applies a localized graph convolution operation to aggregate information from a node's neighbors, effectively capturing both node features and graph structure. This breakthrough enabled efficient representation learning on graphs, leading to significant advances in node classification, link prediction, and graph classification tasks. Over the years, the expressive power of GCN has been constantly studied. Xu et al. (Xu et al. 2019) proposed a theoretical framework to analyze the expressive power of GNNs and designed the Graph Isomorphism Network (GIN) model, which matches the representational capacity of the Weisfeiler-Lehman (WL) (Weisfeiler and Leman 1968) graph isomorphism test.

In this paper, we develop a GIN-based network alignment model and topological enhancement. Unlike previous techniques, our model provides theoretical support for both model selection and alignment refinement.

6 Conclusion

We proposed IGA, an unsupervised framework for anchor link prediction under structural inconsistency and data incompleteness. Built upon the assumption that source and target networks are sampled from a shared random process, IGA integrates three key components: (1) consistency, achieved via GIN-based structural encoding; (2) disparity, introduced through Laplacian reconstruction to regulate intra-network similarity; and (3) robustness, ensured by adaptive refinement through topological augmentation and semantic weighting. Extensive experiments validate the effectiveness and resilience of IGA across diverse benchmarks, with visualization analyses highlighting the progressive identification of backbone correspondences.

Table 5 Statistical details of networks

Networks	#Nodes	#Edges	#Attribute	Avg. deg
Douban online	3906	8164	538	4.2
Douban offline	1118	1511	538	2.7
Allmovie	6011	124709	14	41.4
Imdb	5713	119073	14	41.7
Cora	2708	5278	20	3.9
Citeseer	3312	4732	20	2.9

While this work assumes a shared generative process across networks, the underlying stochastic mechanisms remain insufficiently understood. Future work will explore more expressive models of network generation to provide stronger theoretical guarantees and extend the applicability of unsupervised alignment.

Appendix 1: Dataset details

For completeness, we provide detailed descriptions and statistics of the datasets used in our experiments.

- *Douban online versus Douban offline*: A Chinese social platform where nodes represent users and edges represent friendships. The dataset includes 1,118 anchor links aligning users between online and offline networks (Yin et al. 2018).
- *Allmovie versus Imdb*: Two movie networks where two films are connected if they share at least one actor. Allmovie is derived from Rotten Tomatoes,¹ while Imdb comes from.² The alignment is based on film identity, containing 5,176 anchor links (Trung et al. 2020).
- *Cora*: A citation network with 2,708 scientific papers as nodes and 5,278 citation links (Yang et al. 2016).
- *Citeseer*: A citation network containing 3,312 papers and 4,732 citation links (Yang et al. 2016). For synthetic datasets, the source and target networks are derived by randomly removing a percentage of edges from the original networks to introduce structural noise while keeping node identity unchanged. All node features are reduced to 20 dimensions via truncated SVD decomposition (Halko et al. 2011) (Table 5).

Appendix 2: Proof

Proposition 1 *If source network G_s and target network G_t are isomorphic, then corresponding nodes of anchor links will have the identical label after K iterations of Weisfeiler-Lehman Relabeling.*

¹ <https://www.kaggle.com/ayushkalla1/rotten-tomatoes-movie-database>.

² <https://www.kaggle.com/jyoti1706/imdbmoviesdataset>.

Proof Let $G_s = (V_s, E_s)$ and $G_t = (V_t, E_t)$ be two isomorphic graphs with an isomorphism correspondence $\phi : V_s \rightarrow V_t$. Let $\ell^{(0)}(v)$ denotes the initial label of vertex $v \in V_s$ by degree, its counterpart $v' = \phi(v)$ holds the same degree in initialization.

In the base case:

$$\ell^{(0)}(v) = \ell^{(0)}(\phi(v)) = \ell^{(0)}(v'), \quad \forall v \in V_s, v' \in V_t.$$

Inductive Step: Assume that after k iterations:

$$\ell^{(k)}(v) = \ell^{(k)}(\phi(v)) = \ell^{(k)}(v'), \quad \forall v \in V_s, v' \in V_t.$$

Since G_1 and G_2 are isomorphic, the neighbors satisfy:

$$N(v) \cong N(\phi(v)).$$

Thus, for the update:

$$\begin{aligned} \ell^{(k+1)}(v) &= \text{hash} \left(\ell^{(k)}(v), \{ \ell^{(k)}(u) : u \in N(v) \} \right), \\ \ell^{(k+1)}(\phi(v)) &= \text{hash} \left(\ell^{(k)}(\phi(v)), \{ \ell^{(k)}(z) : z \in N(\phi(v)) \} \right). \end{aligned}$$

Since the neighborhood structure is preserved, we have:

$$\ell^{(k+1)}(v) = \ell^{(k+1)}(\phi(v)).$$

By induction, after K iterations:

$$\ell^{(K)}(v) = \ell^{(K)}(\phi(v)), \quad \forall v \in V_1.$$

Lemma 1 *Initializing the Weisfeiler-Lehman (WL) relabeling process with node-specific feature vectors increases its expressive power in distinguishing non-identical nodes within the same graph.*

Proof Let $G = (V, E)$ be a graph with node set V and edge set E . The 1-WL test assigns initial labels $\ell^{(0)}(v)$ to each node $v \in V$. In the standard setting, $\ell^{(0)}(v)$ is typically uniform or based on simple structural properties (e.g., degree), which may not be sufficient to differentiate nodes with identical local structures.

Now consider initializing $\ell^{(0)}(v)$ with node-specific features $x_v \in \mathbb{R}^d$. These features encode additional semantic information that may not be captured by structure alone. During the WL relabeling process, the updated label at iteration $k + 1$ is given by:

$$\ell^{(k+1)}(v) = \text{HASH} \left(\ell^{(k)}(v), \{ \ell^{(k)}(u) : u \in N(v) \} \right),$$

where $N(v)$ denotes the set of neighbors of node v , and HASH is an injective function. Since the initial labels now reflect node-specific attributes, the multiset of labels aggregated at each step will more accurately reflect unique neighborhood semantics. As a result, even structurally similar nodes may receive different labels if their features differ, allowing the WL process to better distinguish them.

Therefore, incorporating node features as initial labels enhances the discriminative capacity of the WL test within a single graph.

Proof Without the node feature, the 1-WL test can only initialize all nodes identically. Initializing with node degrees is equivalent to starting with identical initialization followed by one round of the relabeling technique.

However, if introducing node features, the 1-WL test could initialize nodes respectively according to their feature, which means that if two nodes share the same feature, it assigns the same color, different color otherwise. In such cases, if two nodes still share the identical topology information of the computational graph and the nodes are initialized differently, the output could still distinguish them according to the property of the HASH function in the iterative process.

$$\ell^{(k+1)}(v) = \text{hash} \left(\ell^{(k)}(v), \{ \ell^{(k)}(u) : u \in N(v) \} \right). \quad (13)$$

Proposition 2 *If two nodes $v \in G_s$, $v' \in G_t$ share isomorphic neighborhoods and identical features under recursive matching, then $H_s^{(l)}(v) = H_t^{(l)}(v')$ for any l under shared GIN parameters.*

Proof To show that $H_s^{(l+1)}(v) = H_t^{(l+1)}(v')$ under the given conditions, we use induction on the number of layers .

Base Case ($l = 0$):

$$H_s^{(0)}(v) = X_s(v), \quad H_t^{(0)}(v') = X_t(v').$$

Induction Hypothesis: Assume $H_s^{(l)}(v) = H_t^{(l)}(v')$ for some $l \geq 0$.

Update Step ($l \rightarrow l + 1$): Using the GIN update rule:

$$\begin{aligned} H_s^{(l+1)}(v) &= \text{MLP} \left((1 + \epsilon) H_s^{(l)}(v) + \sum_{u \in N(v)} H_s^{(l)}(u) \right), \\ H_t^{(l+1)}(v') &= \text{MLP} \left((1 + \epsilon) H_t^{(l)}(v') + \sum_{u' \in N(v')} H_t^{(l)}(u') \right). \end{aligned}$$

By the induction hypothesis and topological consistency:

$$H_s^{(l)}(u) = H_t^{(l)}(u') \implies \sum_{u \in \mathcal{N}(v)} H_s^{(l)}(u) = \sum_{u' \in \mathcal{N}(v')} H_t^{(l)}(u').$$

And the encoding process is in the shared GIN model. By induction, we conclude:

$$H_s^{(l+1)}(v) = H_t^{(l+1)}(v').$$

Proposition 3 *Training to reconstruct the adjacency matrix can introduce hard negatives, degrading anchor link prediction performance.*

Proof Given the adjacency matrix A , if two nodes v and u on source graph are connected (which means $A_s(v, u) = 1$). According to Eq. (3), if trained properly, node embeddings $H_s(v)$ and $H_s(u)$ hold:

$$H'_s(v) \cdot H'_s(u)^T = 1.$$

Since both $H'_s(v)$ and $H'_s(u)$ hold that they have the same norm as 1, so it means $H'_s(v) = H'_s(u)$. If nodes v has its counterpart v' in target graph, according to Eq. (4), the alignment score of them is:

$$S_{vv'} = H'_s(v) \cdot H'_t(v')^T.$$

However, since $H'_s(v) = H'_s(u)$, which means:

$$S_{uv'} = H'_s(u) \cdot H'_t(v')^T = H'_s(v) \cdot H'_t(v')^T = S_{vv'}.$$

Thus the alignment process cannot distinguish node v and u from the same graph, which means it generates “hard negative” node u for v . $\square$

Lemma 2 *Using the normalized Laplacian as reconstruction target mitigates the formation of hard negatives in the downstream alignment task.*

Proof Given the normalized laplace matrix L' , and given two nodes u and v in the same graph, if they are connected, then according to the definition of L' , it holds:

$$L'(u, v) = -\frac{1}{\sqrt{d_u d_v}},$$

where d_u, d_v stands for the degree of node u and v respectively. According to Eq. (3), if trained properly, node embeddings $H_s(v)$ and $H_s(u)$ hold:

$$H'_s(v) \cdot H'_s(u)^T = -\frac{1}{\sqrt{d_u d_v}}.$$

Since both $H'_s(v)$ and $H'_s(u)$ hold that they have the same norm as 1, which means:

$$\cos\theta_{uv} = \frac{H'_s(v) \cdot H'_s(u)^T}{\|H'_s(v)\| \cdot \|H'_s(u)\|} = H'_s(v) \cdot H'_s(u)^T < 0.$$

This means they spread out in embedding space.

Proof Xu et al. (Xu et al. 2019) proved that GIN is as powerful as the 1-WL test in distinguishing graph structures, by demonstrating that its update rule simulates the WL relabeling process. Specifically, in each layer of GIN, node embeddings are updated as follows:

$$h_v^{(k+1)} = \text{MLP}^{(k)} \left((1 + \epsilon) \cdot h_v^{(k)} + \sum_{u \in N(v)} h_u^{(k)} \right),$$

where $h_v^{(k)}$ denotes the embedding of node v at the k -th layer, ϵ is a learnable or fixed scalar, and $\text{MLP}^{(k)}$ is a multilayer perceptron. This formulation effectively replaces the non-learnable HASH function in the WL test with a learnable MLP and additive aggregation.

Given the result in Lemma 1, where node-specific features enhance the expressivity of the WL test, the same benefit extends to GIN, which uses these features as initial embeddings. Furthermore, GIN enables gradient-based optimization, making it more suitable for practical learning tasks such as unsupervised anchor link prediction.

Thus, GIN provides a learnable, expressive, and theoretically grounded alternative to the WL relabeling process.

Acknowledgements This work was supported by the National Natural Science Foundation of China (Grant Nos. 62276013, 62141605, 62050132), the Beijing Natural Science Foundation (Grant No. 1192012), and the Fundamental Research Funds for the Central Universities.

Author contributions Qiming Yang wrote the main text of the manuscript, conducted the experiments, and analyzed the data. Wei Wei contributed to writing and supervised the model design. Ruizhi Zhang conducted the experiments. You Lv contributed to data preprocessing and visualization. Chao Li reviewed the manuscript. Xiangnan Feng provided guidance on model design and data analysis.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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